

WUDSN

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Introduction. For Datafest 2026, WUDSN focused on Hospital Encounter durations. Utilizing the encounters dataset, we isolated "Hospital Encounters" and derived the Length of Stay (LOS) by calculating the difference between the AdmissionsInstant and DischargeInstant timestamps. Optimizing LOS is critical in patient experience and serves as a vital lever for capacity management, patient safety, and economic efficiency. Our research hypothesized that seasonality, specifically the flu window, acts as a primary driver for increased LOS. To evaluate this, we categorized the dataset into Flu season (October - March) and Non-Flu season (April - September) cohorts. Our secondary inquiries expanded on this baseline to quantify how diagnostic categories (indexed by ICD-10 chapters) and departmental assignments further moderate LOS.

Models. We employed a Generalized Linear Model with a Gamma distribution and a log link function to evaluate the determinants of LOS. Our primary analysis revealed a statistically significant seasonal effect ($p < 0.001$), with patients during the non-flu season experiencing a 3% reduction in total stay duration compared to those admitted during the flu season. Furthermore, an Analysis of Deviance confirmed that both diagnostic category (ICD-10 Chapter) and departmental assignment are highly significant predictors of LOS ($p < 0.001$).

Diagnosis. To evaluate diagnostic characteristics across both monthly and flu seasonal scales, we categorized primary diagnoses into 22 distinct chapters based on the ICD-10-CM classification system. This process involved a data-cleaning phase to account for non-standard clinical descriptors and incomplete records, ensuring that each hospital encounter was accurately mapped to its respective diagnostic chapter. We established a baseline for stay durations by calculating the mean LOS for each chapter, which served as a reference point for individual encounter variances. A comparative analysis, visualized via an overlapping bar chart, revealed that LOS was elevated during the flu season for 15 out of 22 chapters (excluding unknown classifications).

Department. To evaluate the role of the department on LOS, we integrated the encounter and department datasets via a merge on DepartmentKey, applying a minimum threshold of 10 encounters. Applying the same seasonal stratification used in our diagnostic analysis, we visualized the variance in LOS by department, highlighting how the "seasonal penalty" of the flu window fluctuates across different specialized clinical environments.

Predictive Models. To model and predict LOS, we evaluated four machine learning architectures – Random Forest, Linear Regression, XGBoost(ensemble learnings) and Single Decision Tree– to identify key determinants. We utilized a mix of categorical and numerical variables, including admission timing, visit classification and the primary drivers established in earlier analysis (ICD-10 chapter, department, and flu seasonality). Among these, the XGBoost model demonstrated the highest predictive accuracy, achieving an R-square of 0.4716.